HEY -WEL MECHANICAL CO.,LTD



Coarse/Fine Bubble Aeration Diffuser



RCD

Fine Bubble Disc Diffuser



Specification

Model	Flow rate (m^3/hr)	Design flow (m^3/hr)	Pressure loss (Pa)	Range depth (m)	Service area (m^2)	Pores quantity	Bubble size (mm)	Max. Temp.	Connector threaded	Weight (kg)
RCD-270	1 – 6	2.0 – 6.0	2000 – 4800	1.5 – 7.0	0.5	6500	1 – 3	100 °C	3/4" NPT	0.6
RCD-350	1 – 12	4.0 – 8.5	3800 – 6300		0.8	12500				0.85

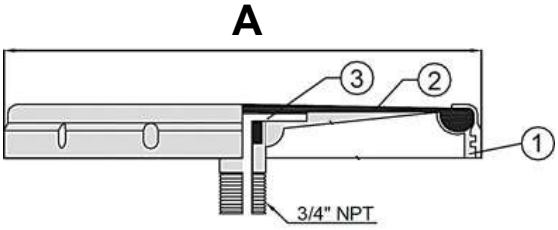
Features

- High Oxygen Transmission Efficiency (OTE) with a large number of identical diameter slits (pores) on EPDM membrane producing tiny air bubbles of an average 1-3 mm, thus provide power saving efficiency and low operation cost.
- Excellent back flow prevention with check valve to avoid clogging and sewage water to entering into aerator pipe line when air transmission is shut (also available with non-check valve).
- Equipped with EPDM membrane with high tensile strength, anti-tearing, and non- deformability. The membranes surface are overspread with over 6500 to 12500 of micro pores. While operating, the bubbles are rise in a form of spiral whirlpool, thus provide the self-cleaning ability for the membranes as well as extended long service life.
- Unique seal ring provide a maximum tight sealing between diffuser base and membrane washer.
- PVC Soft connector for installation with pipe line also available as optional accessories.

Application

- Aeration
- SBR reaction basin
- Sludge activated process
- Wastewatertreatmentprocess
- Denitrification aerobic process
- Aquafarm and other aeration application

Construction



Dimension - A (mm)

RCD-270	RCD-350
270	350

Material

No.	Description	Model	
		RCD270	RCD350
1.	Retaining ring	*PP	*PP
2.	Membrane	**EPDM	**EPDM
3.	Disc	*PP	*PP

*Polypropylene **Ethylene Propylene Diene Monomer *** Not applicable

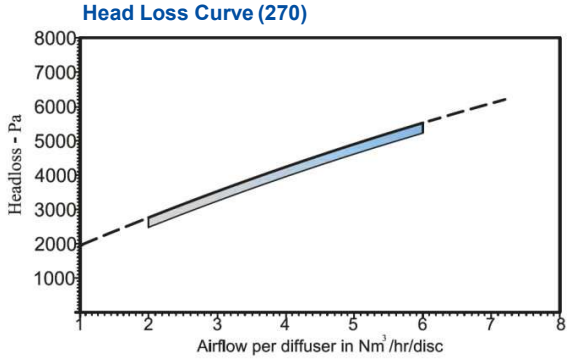
Model: 270

Model: 350

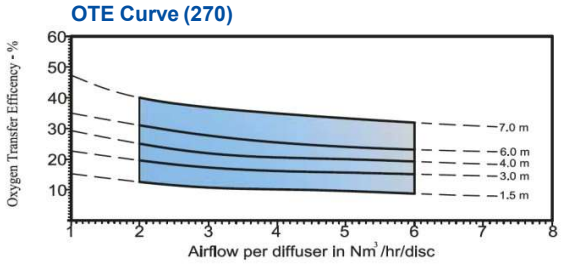


Performance Curve

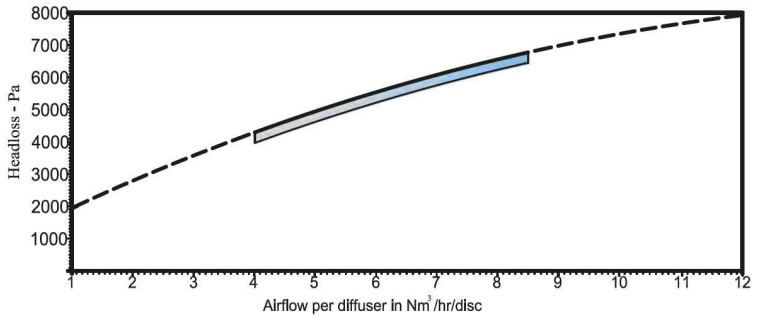
Head Loss Curve (270)



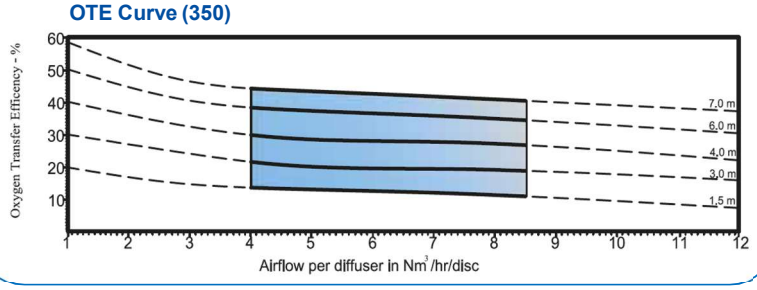
OTE Curve (270)



Head Loss Curve (350)



OTE Curve (350)

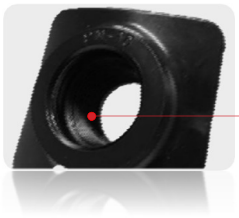


Soft Connector



Ø 32 (Outer dia.)

Oversized pin of Ø 37 mm and curve designed body to provide more secure when installing to the pipeline.

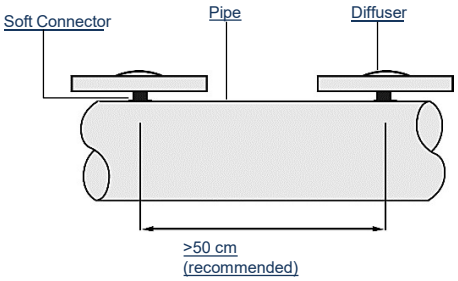


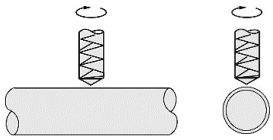
Clockwise threaded for attaching with the diffuser.

Adaptor size	Threaded Dia.	Material
Ø32 mm x 5 mm	1" NTP (Male)	PVC*

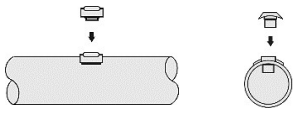
*Polyvinyl Chloride

Installation with Soft Connector

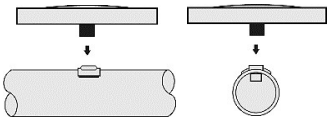




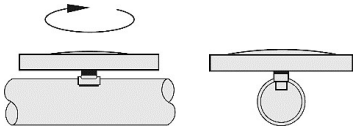
1. Drill a hole of 32 mm diameter.



2. Place the connector into the drilled pipe hole.*



3. Press and rotate (clockwise) the diffuser into the connector.



4. Tightly winding the diffuser by hand and tool when it reach 80% fully fit.

*It is recommended to apply a silicone glue between the connector and the pipe hole for more stable adhesive.

Moister Purge System

As air temperature become higher after compressed by the heat blast of the blower (approximately up to 50 c°), the accumulation of water or liquid will occur inside the aeration pipeline, thus decreasing of oxygen transfer rate and clogging of diffuser's micro pores. In order to prevent these issue, the moisture purge system can provided the benefit of discharging condensed wastewater and liquid from the aeration pipeline which might entered into the pipe by accumulated or when

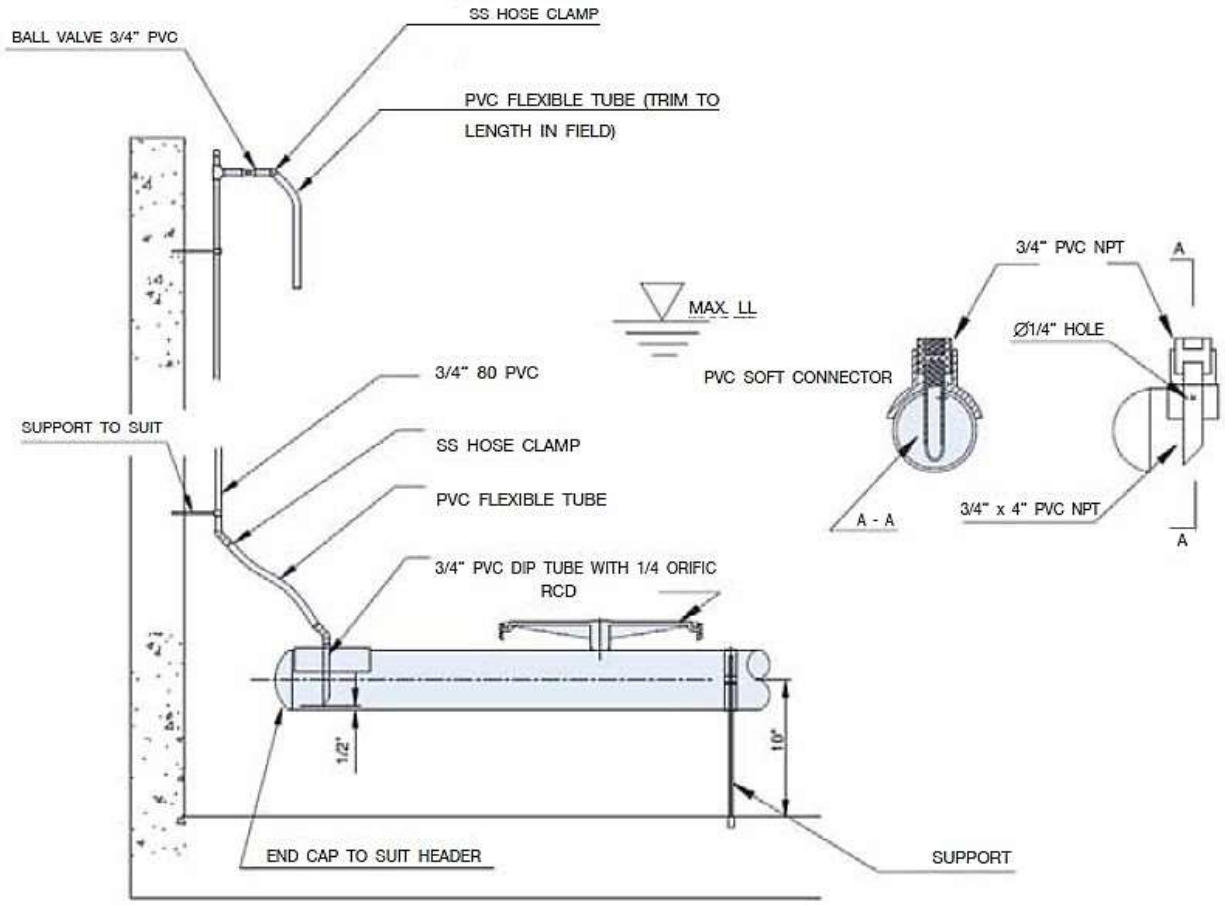
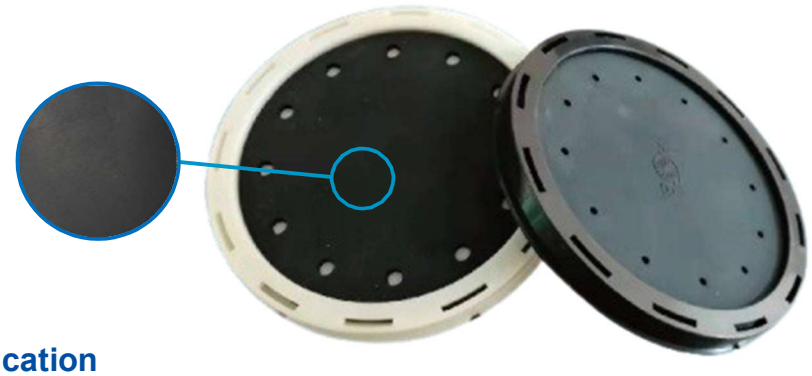
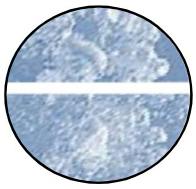


Figure 1. Moister Purge System

Coarse Bubble Disc Diffuser





Specification

Model	Material	Connector threaded	Flow rate	OTE	Head loss	Weight (kg)
RSD-168	-Rubber cap: EPDM* -Main body: ABS**	3/4" NTP (Male)	6– 24 m^3/hr	17% @Water depth 4 m	1550 Pa @ 15 m^3/hr	0.13

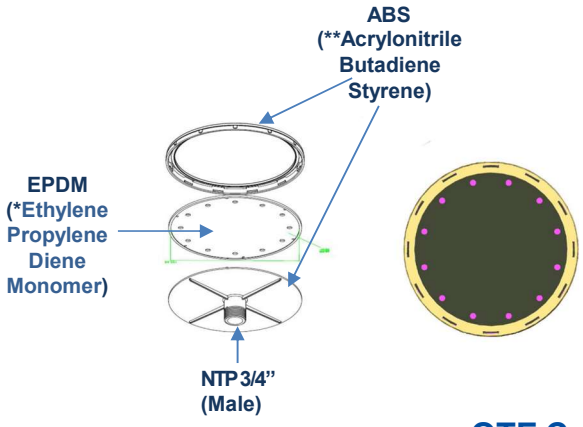
Features

- Special design structure to prevent water back-flow during air flow interruptions.
- Main body made of high quality ABS material with heat resistance and softening temperature prevention.
- Polyurethane membrane offers a durability protection from corrosion, weather, oil, and deformation.
- Adjustable discharge hole for different air volume, which decreases the pressure loss and prevent clogged as well as low maintenance.
- Easy installation with capability to install in upwards or downwards position.

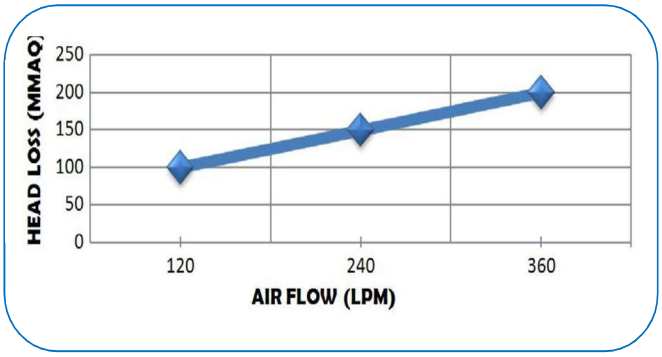
Application

- Aeration
- Wastewater treatment
- Sludge activated process
- Purifywatertreatmentprocess
- Mixing application
- Equalization process

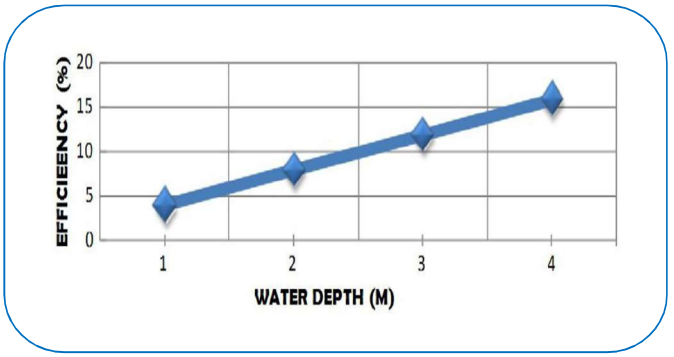
Construction



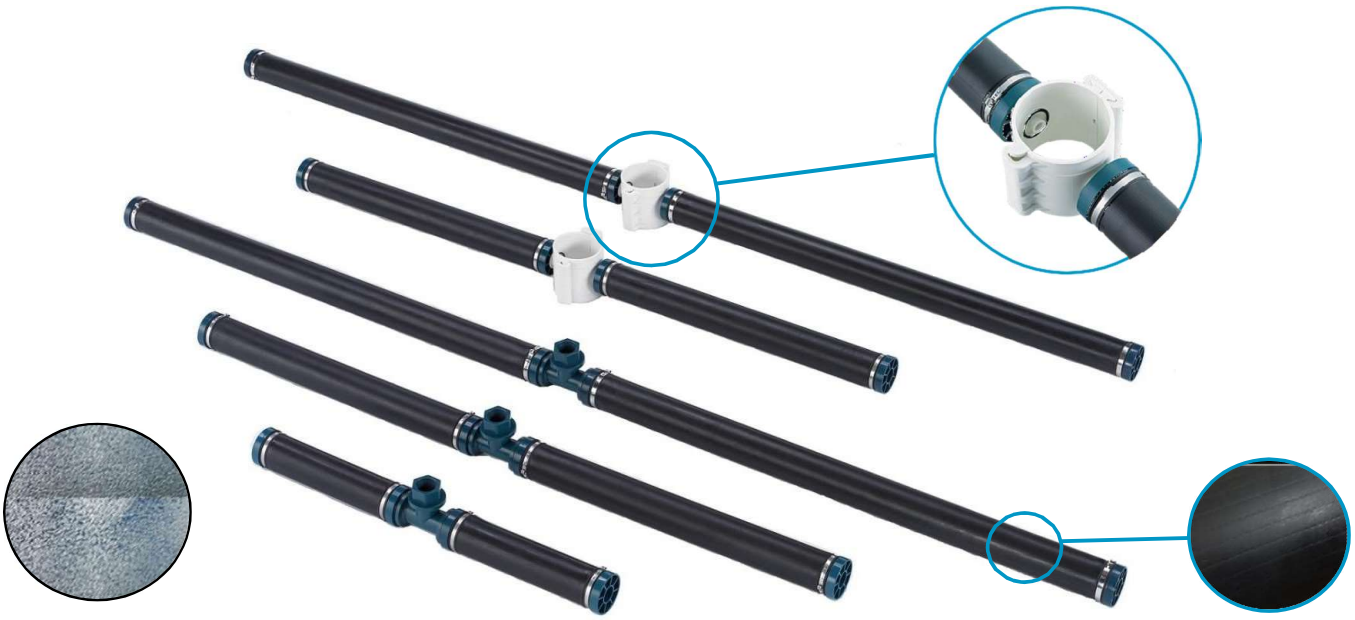
Head Loss Curve



OTE Curve



Fine Bubble Tube Diffuser



Specification

Description / Model		RCT300	RCT600	RCT1000
Material :	Membranes sleeve	EPDM (Ethylene Propylene Diene Monomer)		
	Supported tube	Hard ABS, Open-end designed for buoyancy force minimize.		
Diameter x Perforation length		63 mm × 300 mm	63 mm × 600 mm	63 mm × 1000 mm
Connector-air inlet threaded		R.3/4" PT, Female		
Flow rate : Best continuous operation		10 ~ 80 L/min	30 ~ 250 L/min	50 ~ 320 L/min
Bubble size		1 – 3 mm		

Features

- RCT Tube Diffuser equipped with compression molded EPDM membranes, able to endure against an ultraviolet degradation or chemical corruption.
- Hollow type tube designed allow for maximum strength at a minimum positive buoyancy.
- Tube inlet distributor is designed for all in one injection.
- The supported tube are made of ABS material.
- Membranes are directly fixed by stainless steel (SUS304) clamp on the tube.

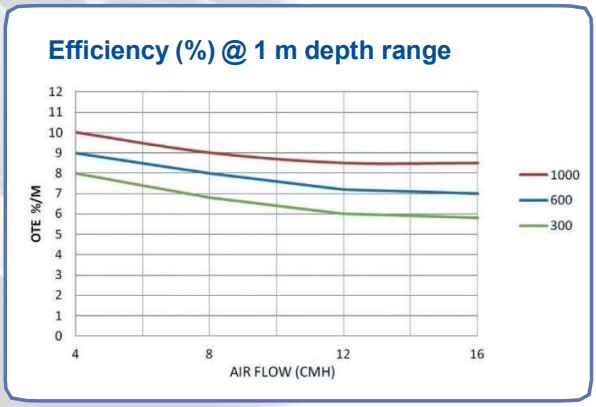
Application

- Municipal sewage treatmentprocess
- Industrial wastewater treatment process
- Purifywatertreatmentprocess
- Ozone diffuser application
- Sludge stabilization process
- Aquafarm

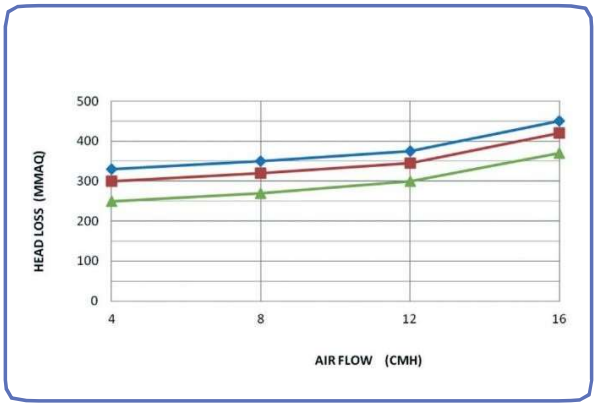
Advantages

- Highoxygentransferefficiencywithlowhead loss.
- Self-cleaning ability and good seal to prevent water fromenter into the pipeline.
- High service area and low power consumption to provide costefficiency.
- Hightensilestrength EPDMmembranematerialwith non- deformability for long-term operation of diffuser.
- Easy installation with ABS adaptor which can be assemblyontopipe withoutusinganyspecialtools.

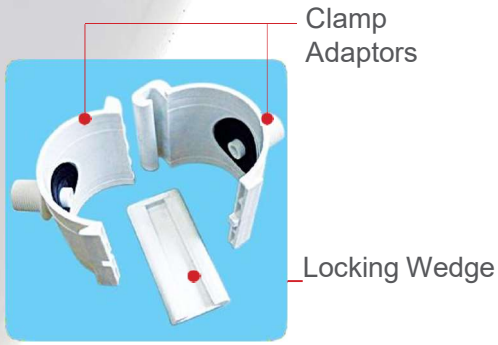
OTE Curve



Head Loss Curve



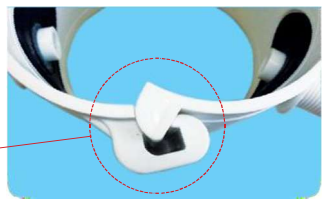
Tube Diffuser Adaptor



Hammer the wedge to lock the clamp tight.



Hook lock design clamp.



Model	Size	Threaded Dia.	Material
CLM-02	Ø90 mm x 4 mm	3/4" (PT)	ABS*

*Acrylonitrile Butadiene Styrene

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